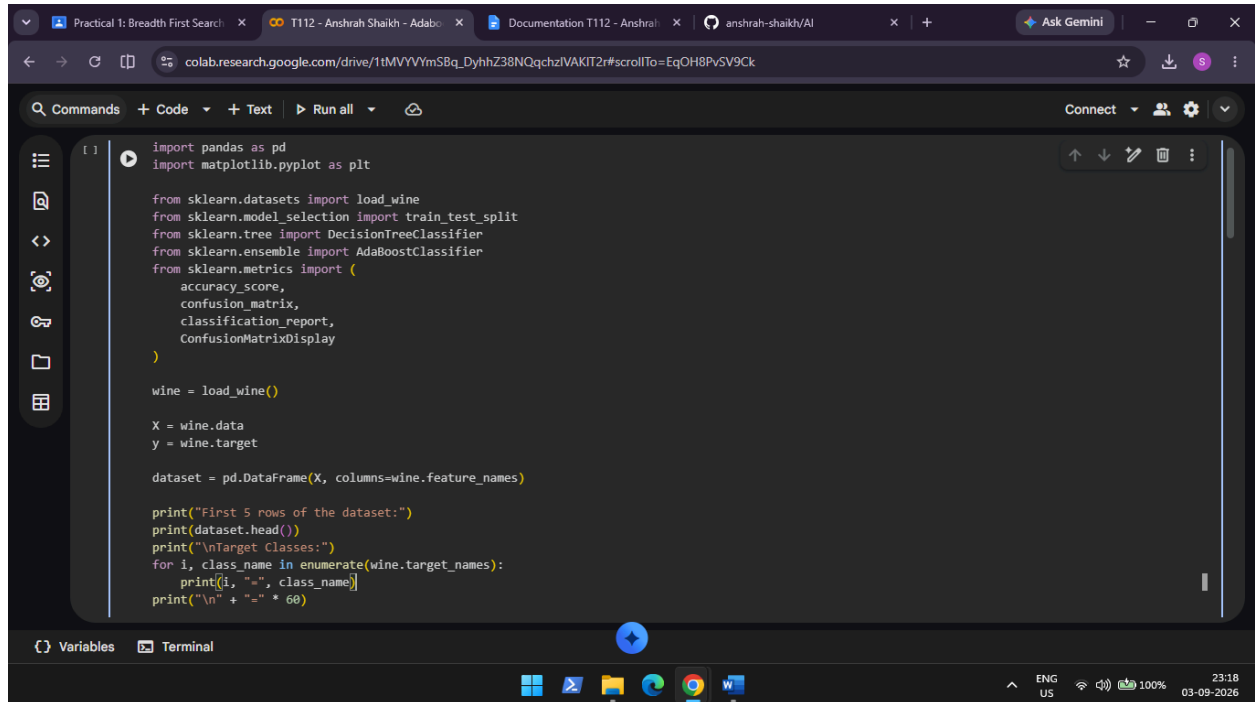


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AIM: Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

INPUT:



```
import pandas as pd
import matplotlib.pyplot as plt

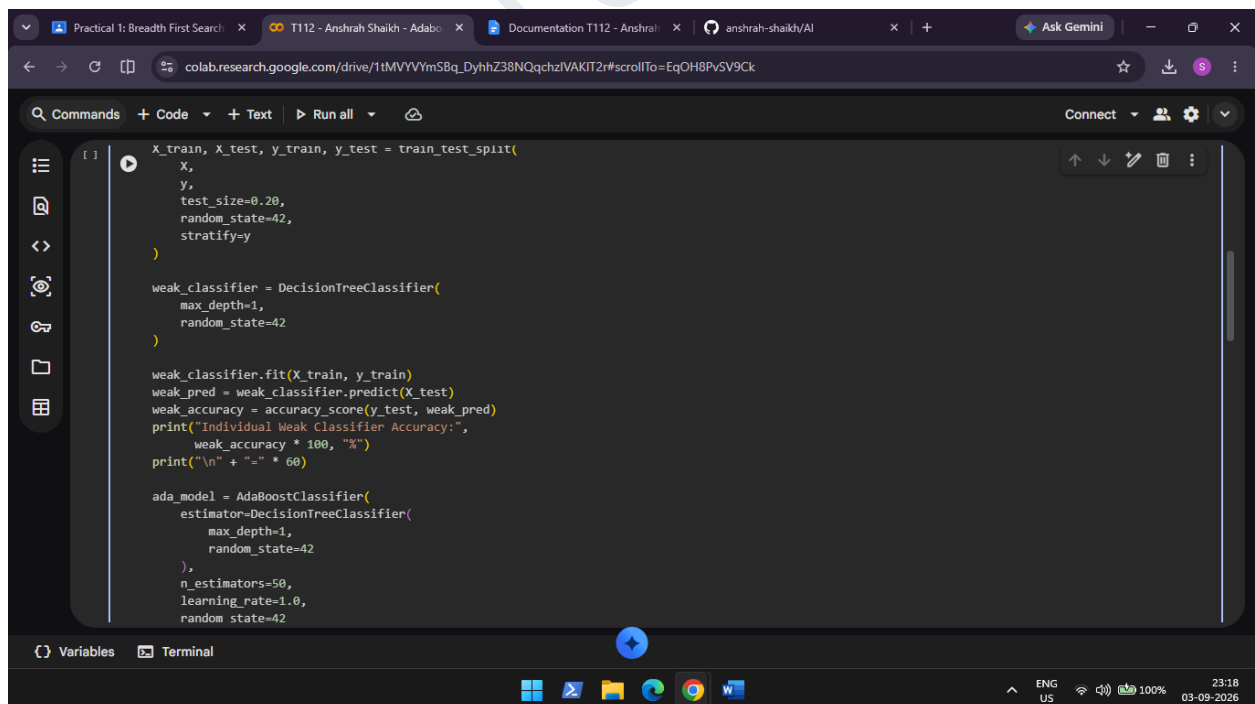
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

wine = load_wine()

X = wine.data
y = wine.target

dataset = pd.DataFrame(X, columns=wine.feature_names)

print("First 5 rows of the dataset:")
print(dataset.head())
print("\nTarget Classes:")
for i, class_name in enumerate(wine.target_names):
    print(i, "-", class_name)
print("\n" + "-" * 60)
```



```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_classifier.fit(X_train, y_train)
weak_pred = weak_classifier.predict(X_test)
weak_accuracy = accuracy_score(y_test, weak_pred)
print("Individual Weak Classifier Accuracy:",
      weak_accuracy * 100, "%")
print("\n" + "-" * 60)

ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)
```

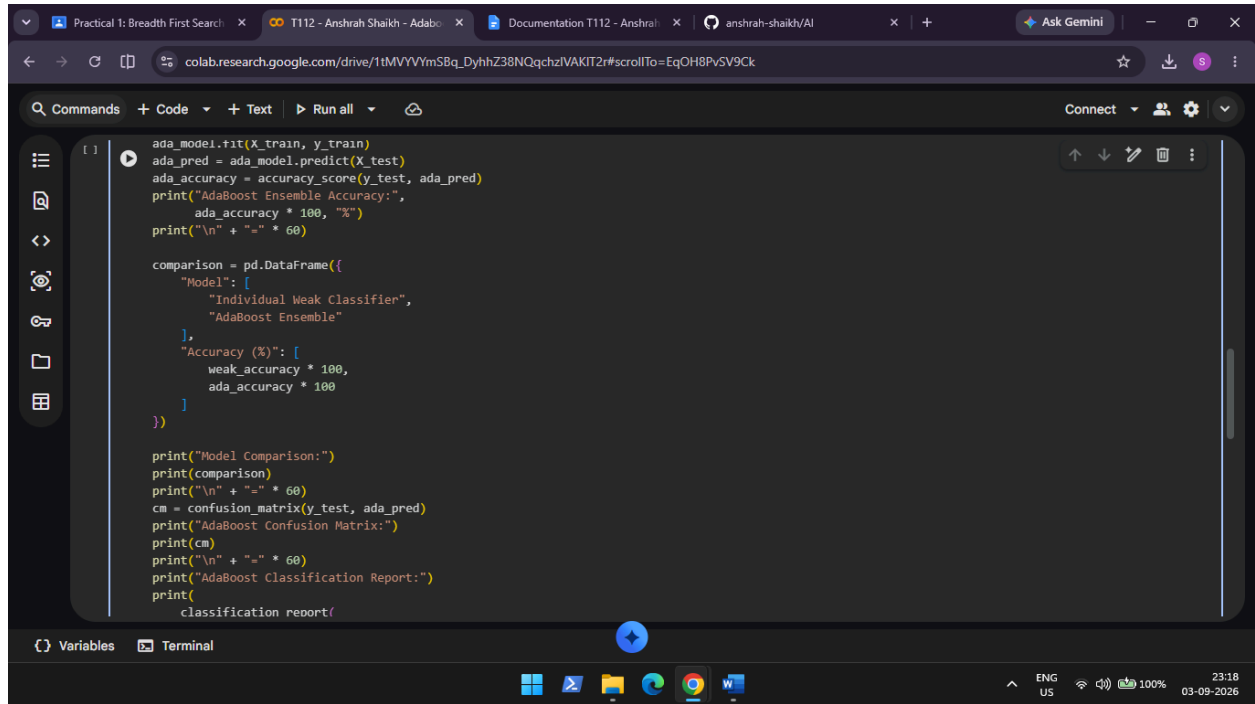
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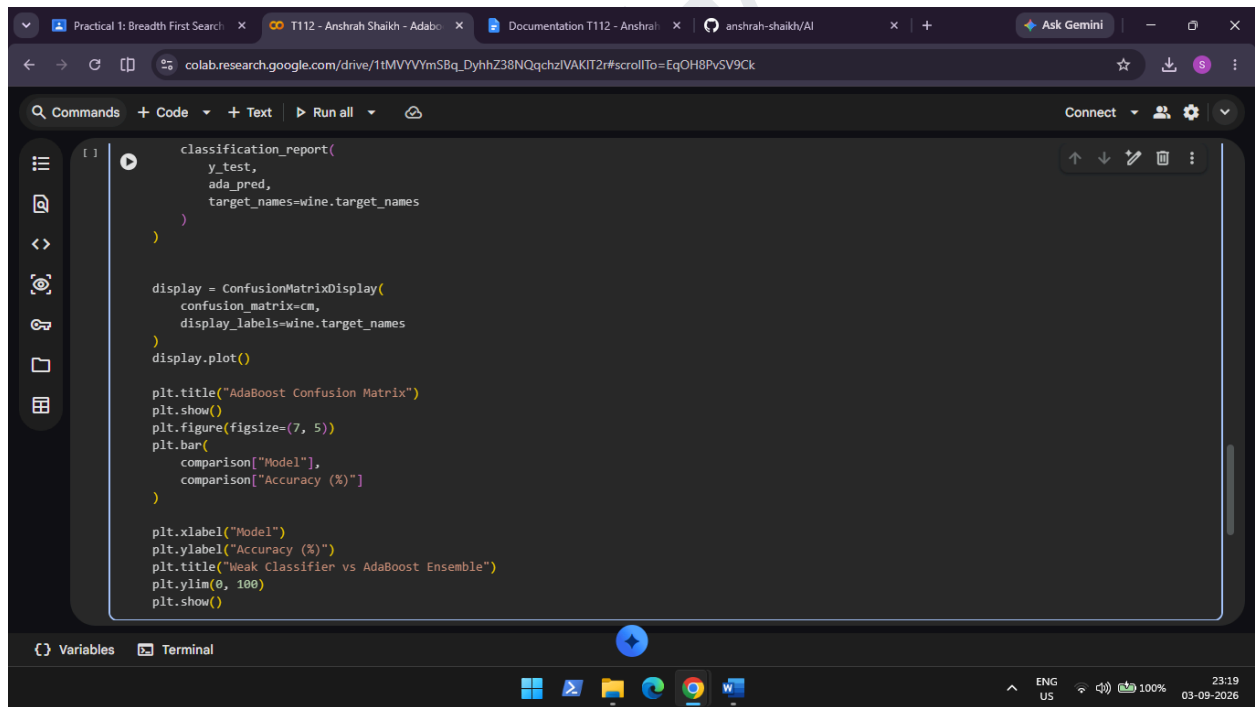
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```
ada_model.fit(X_train, y_train)
ada_pred = ada_model.predict(X_test)
ada_accuracy = accuracy_score(y_test, ada_pred)
print("AdaBoost Ensemble Accuracy:",
      ada_accuracy * 100, "%")
print("\n" + "=" * 60)

comparison = pd.DataFrame({
    "Model": [
        "Individual Weak Classifier",
        "AdaBoost Ensemble"
    ],
    "Accuracy (%)": [
        weak_accuracy * 100,
        ada_accuracy * 100
    ]
})

print("Model Comparison:")
print(comparison)
print("\n" + "=" * 60)
cm = confusion_matrix(y_test, ada_pred)
print("AdaBoost Confusion Matrix:")
print(cm)
print("\n" + "=" * 60)
print("AdaBoost Classification Report:")
print(classification_report(
```



```
classification_report(
    y_test,
    ada_pred,
    target_names=wine.target_names
)

display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=wine.target_names
)
display.plot()

plt.title("AdaBoost Confusion Matrix")
plt.show()
plt.figure(figsize=(7, 5))
plt.bar(
    comparison["Model"],
    comparison["Accuracy (%)"]
)

plt.xlabel("Model")
plt.ylabel("Accuracy (%)")
plt.title("Weak Classifier vs AdaBoost Ensemble")
plt.ylim(0, 100)
plt.show()
```

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OUTPUT:

```
First 5 rows of the dataset:
alcohol  malic_acid  ash  alkalinity_of_ash  magnesium  total_phenols  \
0      14.23      1.71  2.43          15.6        127.0         2.80
1      13.20      1.78  2.14          11.2        100.0         2.65
2      13.16      2.36  2.67          18.6        101.0         2.80
3      14.37      1.95  2.50          16.8        113.0         3.85
4      13.24      2.59  2.87          21.0        118.0         2.80

flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity  hue  \
0          3.06                0.28             2.29           5.04  1.04
1          2.76                0.26             1.28           4.38  1.05
2          3.24                0.30             2.81           5.68  1.03
3          3.49                0.24             2.18           7.80  0.86
4          2.69                0.39             1.82           4.32  1.04

od280/od315_of_diluted_wines  proline
0                3.92        1065.0
1                3.40        1050.0
2                3.17        1185.0
3                3.45        1480.0
4                2.93         735.0

Target Classes:
0 = class_0
1 = class_1
2 = class_2

=====
Individual Weak Classifier Accuracy: 58.33333333333333 %
```

```
=====
Individual Weak Classifier Accuracy: 58.33333333333333 %
AdaBoost Ensemble Accuracy: 91.66666666666666 %

=====
Model Comparison:
      Model  Accuracy (%)
0  Individual Weak Classifier    58.33333
1      AdaBoost Ensemble    91.66667

=====
AdaBoost Confusion Matrix:
[[12  0  0]
 [ 2 12  0]
 [ 0  1  9]]

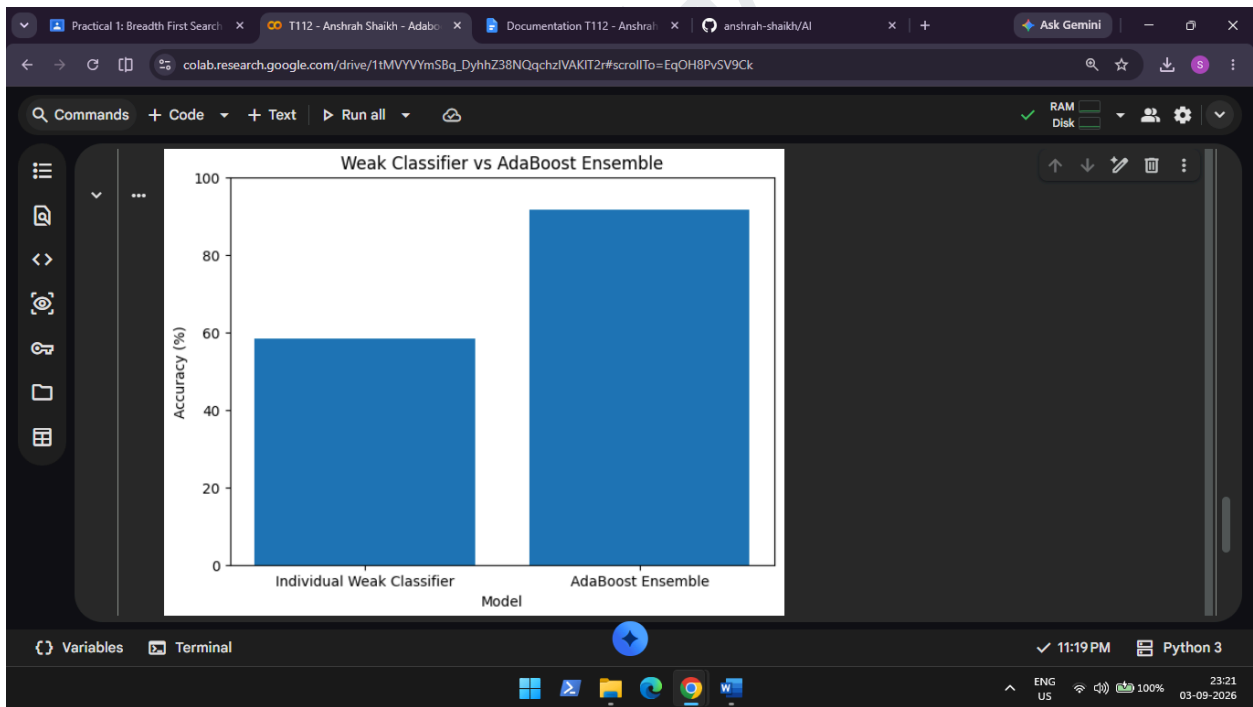
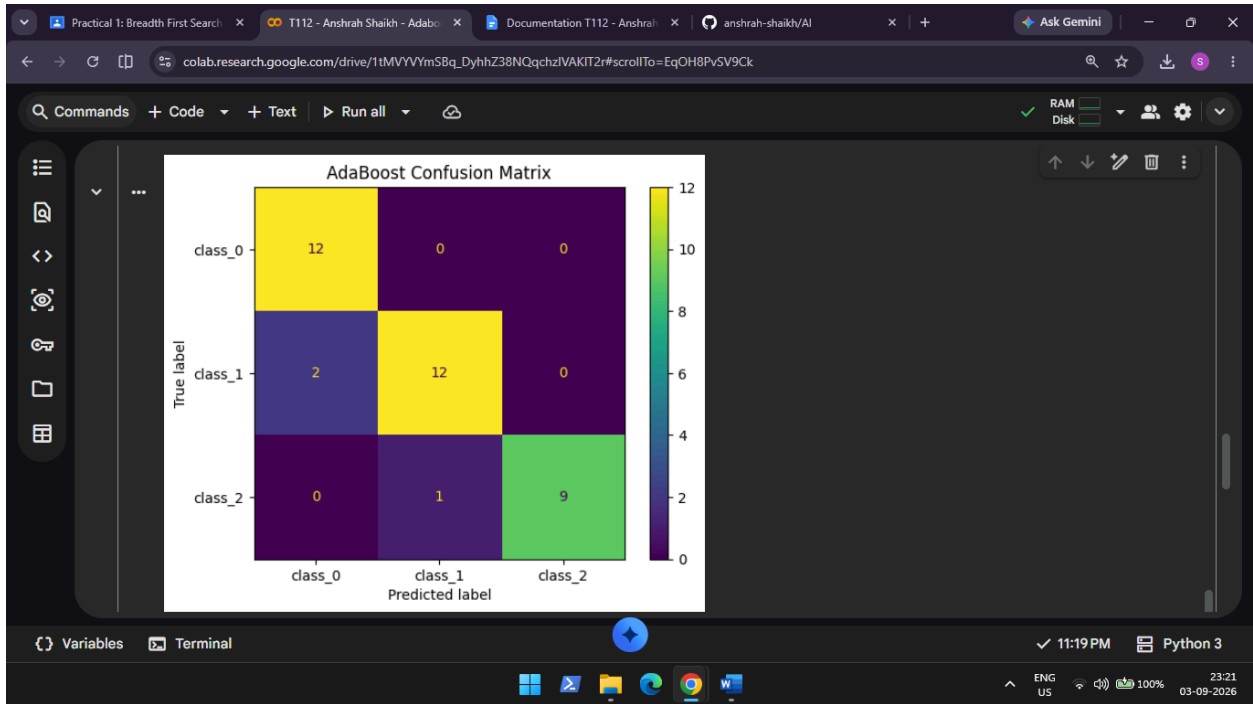
=====
AdaBoost Classification Report:
              precision    recall  f1-score   support

 class_0       0.86       1.00       0.92        12
 class_1       0.92       0.86       0.89        14
 class_2       1.00       0.90       0.95        10

 accuracy              0.92        36
 macro avg       0.93       0.92       0.92        36
 weighted avg     0.92       0.92       0.92        36
```

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